

Software Requirements Specification (SRS): Levisseals Ventures Property Management Platform

Brand Context: Levisseals Ventures / Kevisseal Investments

System Architecture Type: Cloud-Based SaaS Architecture with Multi-Tenant UI Customization and Context-Aware Mobile Routing.

1. Executive System Description

The Levisseals Ventures Property Management Platform is a complete, three-tier cloud ecosystem built to streamline rental management, secure mobile financial transactions, and monetize localized rental communities. The system bridges the gaps between the property manager (Super Admin/App Owner), property owners (Landlords), and residents (Tenants). Unlike traditional management portals, this system features context-driven rebranding based on unique apartment token routing, a robust installment payment framework ("paying in bits"), an on-demand video ad marketing exchange optimized via automated file-size verification, and built-in local micro-farming and civic utilities designed to sustain high daily tenant user retention.

2. Deep-Dive Role Modules & Feature Specifications

A. Tenant Mobile Application Module

The tenant facing app is a native/progressive web application engineered for lightweight installation, long-term token authentication persistence, and localized daily utility usage.

- **Persistent Token Authentication:** Keeps users authenticated safely for 1 to 2 months using secure localized refresh-token loops.
- **Context-Aware UI Customization Engine:** When a user inputs their assigned *Apartment Code* upon registration or unit movement, the app layout, color palette, icons, and official apartment logo dynamically change to match that specific landlord's pre-configured theme.
- **Flexible Rent Hub (Payment in Bits):** Provides a clear progress bar visual of the current month's rent. Accepts asynchronous installment-based mobile money payments, instantly

recalculating and decreasing the outstanding balance on the client view.

- **Historical Document & Archive Lockout:** Historical records (old invoices, payment sheets, prior leases) are protected under a secondary landlord-issued verification code layer for multi-factor data safety.
- **Cross-Property Rent Portability:** Enables a single user account to retain lifetime records of previous tenancies, letting users review historical ledger items even if they relocate to an entirely different property within the platform.
- **The Tenant Ad Exchange (Self-Service Ads):** Built-in video advertising dashboard for tenants running small commercial businesses. Includes a payment gate based on ad parameters (reach size, geofence radius, upload duration) and features an analytics view.
- **"Watch-Reward" Analytics Optimization:** Tenants can unlock visual view-counters and interactive performance metrics of their posted ads completely for free by choosing to watch 2 programmatic platform sponsor ads.

B. Landlord Web & Mobile Portal Module

The landlord module offers property owners control over their digital tenant roster, unique rental financial channels, communication logs, and localized backend styling options.

- **Onboarding & Contract Verification:** Collects verified ID data, email accounts, and mobile coordinates. Supports legal contract signing executed both through in-app digital signature APIs or scanned physical document attachments.
- **Apartment Hierarchy & Code Generator:** Enables landlords to define property structures (blocks, floors, single units) and generate unique, alphanumeric entry keys for tenants.
- **Theme Customization Workspace:** A direct layout editing board where landlords can upload corporate imagery, select corporate brand styles, or enlist the development firm to establish custom themes for a dedicated configuration fee.
- **M-Pesa Gateway Direct Routing Configuration (C2B, B2B, B2C):** Landlords can easily tie their direct business shortcodes (Paybills/Buy Goods Tills) into the application layer for direct Consumer-to-Business (C2B) rent collections. Includes multi-modal triggers to initiate Business-to-Consumer (B2C) payouts or direct vendor payment options (B2B).
- **Milestone Alert Core:** Pushes real-time alerts across SMS, email, in-app notifications, and WhatsApp endpoints whenever a tenant makes a payment or posts an immediate service complaint.
- **Peer-to-Peer Landlord Network:** Integrated chat hub connecting active platform property owners to share community notes, contractor references, or neighborhood updates.

C. Property Manager (Super Admin) Master Dashboard

The master admin tier is the core structural oversight board used by Levisseals Ventures to

monitor platform activity, deploy global parameters, and process system wide financial data.

- **Global System Ledger & Auditing Control:** Complete structural view of all system transactions, monitoring global tenant collections, individual landlord payment distributions, and system platform cuts.
- **Master Gateway Routing Panel:** Administrative controls to adjust API connection details for primary mobile money channels, manage global transaction cuts, and approve mass landlord payout settlements.
- **Ad Network Approval & Allocation Suite:** Global clearance engine to monitor, audit, filter, and schedule third-party ads (like Google AdMob), dedicated local ad videos, or captive portal WiFi splash page configurations.
- **Legal Compliance Repository:** Secure system database hosting all signed contract documents, landlord ID credentials, and legal compliance reports.

3. Platform Integration & System Matrix

The following matrix outlines exactly how workflows translate and cross-update data states across all three system tiers simultaneously:

Workflow Event	Tenant App Action/State	Landlord Portal Action/State	Property Manager Admin State
Onboarding / Unit Matching	Inputs apartment code; UI downloads custom layout styles instantly.	Issues alphanumeric unit code; views tenant onboarding docs.	System maps the relational DB linkage and logs active unit volume.
Installment Payment ("Bits")	Triggers STK push; progress bar drops balance in real time.	Receives instant notification; ledger sheet flags updated balance.	Logs transaction volume; aggregates service fee percentage.
Video Ad Placement	Defines reach criteria; clears video upload payment gate.	Sees ad space allocations within the local tenant community board.	Ad content flags for approval queue; handles processing calculation.

Workflow Event	Tenant App Action/State	Landlord Portal Action/State	Property Manager Admin State
Reward Loop Execution	Watches 2 programmatic ads; unlocks views counter analytics dashboard.	Maintains clean, active, non-obtrusive user viewing sessions.	Registers impressions from ad networks, collecting ad revenue.

4. Value-Added System Extensions

A. Livestock & Micro-Farming Production Ledger

This module provides tenants with small-scale agricultural projects an integrated everyday system utility to manage farm production on premises.

- **Stock Inventory:** Digital counting panels for poultry, goats, dairy cattle, or rabbit configurations.
- **Financial Bookkeeping Tracker:** Simple ledger tool tracking daily livestock feed costs, medicinal purchases, and veterinary fees measured against sales from eggs, milk, or meat.
- **Immunization & Treatment Timelines:** Automated reminder triggers alerting tenants via WhatsApp or SMS updates when upcoming livestock vaccines, dewormings, or diagnostic checkups are due.

B. Chama & Social Savings Collective Toolkit

A communal tool for rotating credit structures or community savings operations managed within local neighborhoods or apartment groupings.

- **Asynchronous Contribution Ledgers:** Real-time balancing logs showcasing individual periodic savings pool contributions, flagging outstanding members.
- **Merry-Go-Round Rotational Engine:** Automatically calculates and updates payout schedules, indicating whose turn it is to receive the communal pool funds.

C. Local Services Directory & Neighborhood Noticeboard

- **Local Contractor Exchange:** Built-in directory connecting tenants with certified local electricians, gas providers, plumbing professionals, or clean-up teams.
- **Tenant Gig Board:** Free ad marketplace allowing residents to market internal personal skills (sewing, home tutoring, baking) right to their neighbors.
- **Real-Time Community Alerts:** Crowdsourced neighborhood news timeline detailing municipal water shortfalls, electricity updates, or gridlock traffic reports.

5. Technical Implementation Blueprint

Developers implementing this platform architecture should look to the following layout guide when setting up APIs and data schemas:

[SYSTEM ROUTING ARCHITECTURE]

